CHAPTER 8

Revision and consolidation

8.1 Revision strategy

> **Key point.** Effective revision alternates retrieval from memory, explicit correction of mistakes and reuse in a problem. Passively rereading the course is not enough: you must attempt a question without a model, compare your answer, then redo the question after a delay.

For each chapter, use the following code: **V** (mastered), **O** (shaky), **R** (to redo). After the diagnosis, work first on an orange or red skill, then check immediately with a similar question.

Skill	V	O	R	Remedial action
Choosing a model	☐	☐	☐	Redo a sorting exercise: arithmetic/geometric/exponential.
Reading and calculating	☐	☐	☐	Redo the targeted fluency exercises.
Writing up and interpreting	☐	☐	☐	Add data, formula, calculation, unit and concluding sentence.

Session 52 — Diagnosis of what has been learned

Revision and consolidation

11 May 2027 • B: 2 h – C: 1 h

Objectives.

- Identify what has been learned.
- Classify the mistakes.
- Plan the remediation.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 25%. $1 + \dfrac{25}{100} = 1.25$
2. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
3. Solve $10^x = 1000$. $x = 3$
4. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
5. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: identify what has been learned and classify the mistakes.

Activity 1. Answer the following 14 questions without notes, two per chapter. Record each error in a grid: knowledge, choice of method, calculation, reading, written presentation. Then choose two personal priorities.

1. Sequences. The sequence (u_n) is arithmetic with first term $u_0 = 7$ and common difference -3. Calculate u_{10}. -23
2. Sequences. The sequence (v_n) is geometric with first term $v_0 = 200$ and common ratio 1.05. Calculate v_2. 220.5
3. Exponential functions. State the direction of variation of $f(x) = 3 \times 0.8^x$ on $[0, +\infty)$. **decreasing**
4. Exponential functions. A quantity of 500 increases by 4% per year. Write the model $f(x)$ after x years. 500×1.04^x
5. Common logarithm. Solve $10^x = 0.01$. $x = -2$
6. Common logarithm. Calculate $\log(1000) - \log(10)$. 2
7. Reciprocal function. Solve $\dfrac{1}{x} = 4$ for $x \neq 0$. 0.25
8. Reciprocal function. State the direction of variation of $x \mapsto \dfrac{1}{x}$ on $(0, +\infty)$. **decreasing**
9. Two variables. With the fit $y = 2x + 3$, estimate y for $x = 10$. 23

10. Two variables. Give the mean point of the series x: 1, 2, 3 and y: 4, 6, 8. $G(2,6)$
11. Conditional probability. $P(A) = 0.4$ and $P_A(B) = 0.5$. Calculate $P(A \cap B)$. 0.2
12. Conditional probability. $P(A \cap B) = 0.12$ and $P(A) = 0.3$. Calculate $P_A(B)$. 0.4
13. Binomial distribution. X follows the distribution $\mathcal{B}(3, 0.5)$. Calculate $P(X = 3)$. 0.125
14. Random variable. X takes the values 0, 1, 2 with probabilities 0.5, 0.3, 0.2. Calculate $E(X)$. 0.7

Solution. The correction expects a profile of mistakes and a plan for catching up, not just an overall mark. Have students say aloud which piece of data triggers the choice of method.

Written record. The diagnosis is used to choose the remediation: recognise the mistake, recall the useful rule, redo a similar question, then check without help.

Example 1. Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Solution. The correction expects a profile of mistakes and a plan for catching up, not just an overall mark.

Exercise 8.1 · essential.

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Solution. Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Application. A cooperative installs home solar kits in a rural region. The table below gives the number of kits installed each year.

Year	2021	2022	2023
Number of kits installed	2,480	3,150	3,938

Part A

1. Calculate the rate of change in the number of kits installed between 2021 and 2022, then between 2022 and 2023 (as a percentage, rounded to 0.1%).
2. Calculate the geometric mean of 1.27 and 1.25 (rounded to 10^{-4}).
3. Deduce the average annual rate of change between 2021 and 2023.

Part B

Assume that from 2023 onwards, the number of kits installed increases by 26% per year. Let u_n be the number of kits installed during the year $2023 + n$, values being rounded to the nearest unit.

1. Give u_0, then calculate u_1 and u_2.
2. What kind of sequence is (u_n)? Express u_n in terms of n.
3. According to this model, how many kits would be installed in 2028?
4. Using a calculator, determine the year from which the annual number of kits installed would exceed 20,000.

Solution. **Target:** Rates of change, geometric mean, geometric sequence.

Part A

1. $\dfrac{3{,}150-2{,}480}{2{,}480}\approx 0.270$, an increase of about 27.0%; $\dfrac{3{,}938-3{,}150}{3{,}150}\approx 0.250$, an increase of about 25.0%. The multipliers are therefore about 1.27 and 1.25.
2. $\sqrt{1.27\times 1.25}=\sqrt{1.5875}\approx 1.2600$.
3. The average annual multiplier is about 1.26: the average annual rate is about 26.0% (direct check: $\sqrt{3{,}938/2{,}480}\approx 1.2601$).

Part B

1. $u_0=3{,}938$; $u_1=3{,}938\times 1.26\approx 4{,}962$; $u_2=3{,}938\times 1.26^2\approx 6{,}252$.
2. An increase of 26% means multiplying by 1.26: (u_n) is geometric with first term $u_0=3{,}938$ and common ratio $q=1.26$, so $u_n=3{,}938\times 1.26^n$.
3. 2028 corresponds to $n=5$: $u_5=3{,}938\times 1.26^5\approx 12{,}506$ kits.
4. From the table of values: $u_7\approx 19{,}855<20{,}000$ and $u_8\approx 25{,}017>20{,}000$. Since the ratio is greater than 1, the sequence is increasing: the threshold is exceeded from $n=8$ onwards, that is, in 2031.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 380 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 53 — Reactivation: exponential functions

Revision and consolidation

13 May 2027 • B: 1 h – C: 2 h

Objectives.

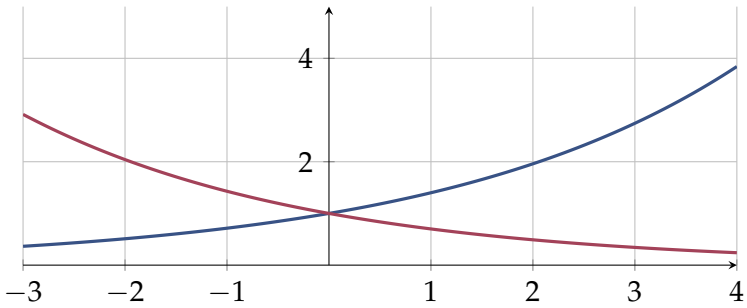
- Reactivate exponential models.
- Solve and interpret.
- Check the units.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 7 - x + 2$. $2x + 9$
2. Solve $10^x = 1000$. $x = 3$
3. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
4. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
5. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate exponential models and solve and interpret.

Activity 2. A value of 18,000 increases by 3.5% per year. Write the model, calculate the value after eight years and determine when it will exceed 25,000.



Solution. $V(t) = 18{,}000 \times 1.035^t$, $V(8) \approx 23{,}703$. The threshold is exceeded from the tenth year onwards. Have students say aloud which piece of data triggers the choice of method.

Written record. An exponential function models a constant relative change. First identify the initial value and the multiplier.

Example 2. Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Solution. $V(t) = 18{,}000 \times 1.035^t$, $V(8) \approx 23{,}703$. The threshold is exceeded from the tenth year onwards.

Exercise 8.2 · essential.

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Solution. Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Application. The number of visitors to a nature park went through three successive annual changes: an increase of 5.4%, then a decrease of 2.6%, then an increase of 7.1%.

1. Calculate the overall multiplier over the three years and the corresponding overall rate of change (rounded to 0.01%).
2. Calculate the average annual rate of change over this period (rounded to 0.01%).
3. The initial number of visitors was 85,000. Calculate the final number of visitors, then check that an increase at this average rate applied three times leads to roughly the same result.

Solution. **Target:** Average annual rate over three years.

1. The multipliers are 1.054, $1 - 0.026 = 0.974$ and 1.071. Overall multiplier: $1.054 \times 0.974 \times 1.071 \approx 1.099\,48$, an overall increase of about 9.95%.
2. The average annual multiplier c_m satisfies $c_m^3 = 1.099\,48$, so $c_m = 1.099\,48^{1/3} \approx 1.0321$. The average annual rate is about 3.21%.
3. Final number of visitors: $85{,}000 \times 1.054 \times 0.974 \times 1.071 \approx 93{,}456$. With the average rate: $85{,}000 \times 1.0321^3 \approx 93{,}451$ visitors; the difference only comes from rounding c_m.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 385 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 54 — Reactivation: common logarithm (base 10)

Revision and consolidation — 18 May 2027 • B: 2 h – C: 1 h

Objectives.

- Reactivate logarithms.
- Solve an equation.
- Check the conditions.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 160. 8
2. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
3. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
4. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
5. Solve $10^x = 1000$. $x = 3$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate logarithms and solve an equation.

Activity 3. Solve $10^x = 340$, $1.06^n = 2$ and $1.06^n \geq 2$. For each answer, distinguish the real value, the approximate value and the minimum integer index.

Solution. $x \approx 2.5315$; $n \approx 11.90$; for the integer threshold, $n = 12$. Have students say aloud which piece of data triggers the choice of method.

Written record. The logarithm turns an exponential equation into an equation in the exponent. Conditions: base positive and different from 1, right-hand side positive.

Example 3. Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Solution. $x \approx 2.5315$; $n \approx 11.90$; for the integer threshold, $n = 12$.

Exercise 8.3 · essential.

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.

3. Calculate a probability from a tree, then interpret it.

Solution. Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Application. *Definition.* The order of magnitude of a real number $x > 0$ is the power of 10 closest to x (compare the distances from x to the two powers of 10 on either side of it).

1. Consider $x = 27.6$ and $y = 812$.
   a. Give the order of magnitude of x and of y, justifying with a calculation of distances.
   b. Using a calculator, give $\log(x)$ and $\log(y)$ rounded to 10^{-3}. Make a conjecture: how can the order of magnitude be obtained from the logarithm rounded to the nearest unit?
2. Consider $z = 4.6$. Determine its order of magnitude and $\log(z)$. Does the conjecture of question 1 hold?
3. Write x in scientific notation: $x = a \times 10^n$ with $1 \leqslant a < 10$ and n an integer. Depending on the value of a, what is the order of magnitude of x? (If the distances are equal, take the larger power.)
4. Express $\log(x)$ in terms of n and $\log(a)$. Deduce a method for obtaining n, a and then the order of magnitude of x from $\log(x)$.
5. In cybersecurity, the strength of a key or a password is assessed by the number of possibilities. A calculator cannot display the following numbers directly. Determine the order of magnitude of:
   a. the number $2^{1,024}$ of 1,024-bit keys;
   b. the number 94^{128} of 128-character passwords chosen from 94 characters.

Solution. **Target:** Order of magnitude and common logarithm.

1. a. $10 < 27.6 < 100$: $27.6 - 10 = 17.6$ and $100 - 27.6 = 72.4$, so the order of magnitude of x is 10. $100 < 812 < 1{,}000$: $812 - 100 = 712$ and $1{,}000 - 812 = 188$, so that of y is 10^3.
   b. $\log(27.6) \approx 1.441$ and $\log(812) \approx 2.910$. Rounded to the nearest unit: 1 and 3. Conjecture: "the order of magnitude of x is 10^k, where k is $\log(x)$ rounded to the nearest unit".
2. $1 < 4.6 < 10$: $4.6 - 1 = 3.6$ and $10 - 4.6 = 5.4$, so the order of magnitude of z is $1 = 10^0$. But $\log(4.6) \approx 0.663$, which rounds to 1: the conjecture would give 10^1. It is false (z is a counterexample).
3. x lies between 10^n and 10^{n+1}, whose midpoint is $\dfrac{10^n + 10^{n+1}}{2} = 5.5 \times 10^n$. So the order of magnitude is 10^n if $a < 5.5$, and 10^{n+1} if $a \geqslant 5.5$.
4. $\log(x) = \log(a) + \log(10^n) = n + \log(a)$, with $0 \leqslant \log(a) < 1$. Method: n is the integer part of $\log(x)$; $a = 10^{\log(x) - n}$; compare a with 5.5 (which amounts to comparing $\log(x) - n$ with $\log(5.5) \approx 0.740$, not with 0.5).
5. a. $\log(2^{1,024}) = 1{,}024 \log(2) \approx 308.255$. So $n = 308$, $a = 10^{0.255} \approx 1.80 < 5.5$: the order of magnitude is 10^{308}.

b. $\log(94^{128}) = 128\log(94) \approx 252.560$. So $n = 252$, $a = 10^{0.560} \approx 3.63 < 5.5$: the order of magnitude is 10^{252}. (Rounding the logarithm would wrongly give 10^{253}.)

Homework. Redo practice exercise 3 with the following data: replace the initial value by 390 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 55 — Reactivation: reciprocal function

Revision and consolidation

20 May 2027 • B: 1 h – C: 2 h

Objectives.

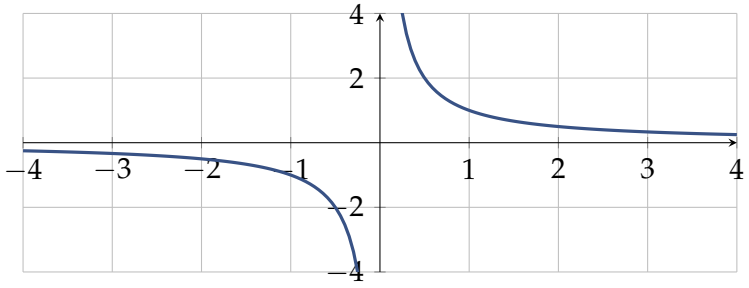
- Reactivate the reciprocal function.
- Carry out a study.
- Sketch and interpret.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 10%. $1 + \dfrac{10}{100} = 1.10$
2. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
3. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
4. Solve $10^x = 1000$. $x = 3$
5. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate the reciprocal function and carry out a study.

Activity 4. Study $f(x) = x + \dfrac{16}{x}$ on $]0, +\infty[$ and interpret its minimum in an average-cost context. Produce the table of variations and a sketch.



Solution. $f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$. Have students say aloud which piece of data triggers the choice of method.

Written record. For a function containing $\dfrac{1}{x}$, start by excluding $x = 0$, then differentiate, factorise and study the sign before the variations.

Example 4. Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Solution. $f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$.

Exercise 8.4 · essential.

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Solution. Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Application. A workshop makes raffia bags. The production cost, in thousands of ariary, of x bags per week is given by $C(x) = 2x^2 + 40x + 1{,}250$, for x between 0 and 60. The workshop is looking for the number of bags for which the average cost of making one bag is lowest.

1. Calculate $C(0)$ and interpret this result. Calculate the production cost of 10 bags, then the average cost of one bag for this production.
2. Let $f(x)$ be the average cost of one bag when the workshop makes x bags, for $x \in [1, 60]$. Show that $f(x) = 2x + 40 + \dfrac{1{,}250}{x}$.
3. Calculate $f'(x)$ and show that $f'(x) = \dfrac{2(x-25)(x+25)}{x^2}$.
4. Study the sign of $f'(x)$ on $[1, 60]$ and draw up the table of variations of f.
5. How many bags should the workshop make for the average cost to be minimal? Give this minimum average cost.
6. The bags are sold for 150,000 Ar each. For this production, what is the profit made on one bag? Over the week?

Solution. **Target:** Minimum average cost: derivative and reciprocal.

1. $C(0) = 1{,}250$: the fixed costs amount to 1,250 thousand ariary, that is, 1,250,000 Ar per week. $C(10) = 200 + 400 + 1{,}250 = 1{,}850$ thousand ariary; average cost: $\dfrac{1{,}850}{10} = 185$ thousand ariary, that is, 185,000 Ar per bag.
2. For $x \geqslant 1$, $f(x) = \dfrac{C(x)}{x} = \dfrac{2x^2}{x} + \dfrac{40x}{x} + \dfrac{1{,}250}{x} = 2x + 40 + \dfrac{1{,}250}{x}$.
3. The derivative of $\dfrac{1}{x}$ is $-\dfrac{1}{x^2}$, so $f'(x) = 2 - \dfrac{1{,}250}{x^2} = \dfrac{2x^2 - 1{,}250}{x^2} = \dfrac{2(x^2 - 625)}{x^2} = \dfrac{2(x-25)(x+25)}{x^2}$.

4. On $[1,60]$: $2>0$, $x+25>0$ and $x^2>0$, so $f'(x)$ has the sign of $x-25$. Thus $f'(x)<0$ on $[1,25)$, $f'(25)=0$ and $f'(x)>0$ on $(25,60]$. f is decreasing on $[1,25]$ and increasing on $[25,60]$, with $f(1)=1{,}292$, $f(25)=140$ and $f(60)\approx 180.8$.
5. The minimum is reached for $x=25$: 25 bags should be made per week. The minimum average cost is $f(25)=50+40+50=140$ thousand ariary, that is, 140,000 Ar per bag.
6. Profit per bag: $150{,}000-140{,}000=10{,}000$ Ar. Over the week: $25\times 10{,}000=250{,}000$ Ar (check: revenue $25\times 150=3{,}750$ and cost $C(25)=3{,}500$ thousand ariary, that is, 250 thousand ariary).

Homework. Redo practice exercise 2 with the following data: replace the initial value by 395 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.